ITDS 124 Mathematics of Computing

Chapter 3. Counting

3.1 Permutations

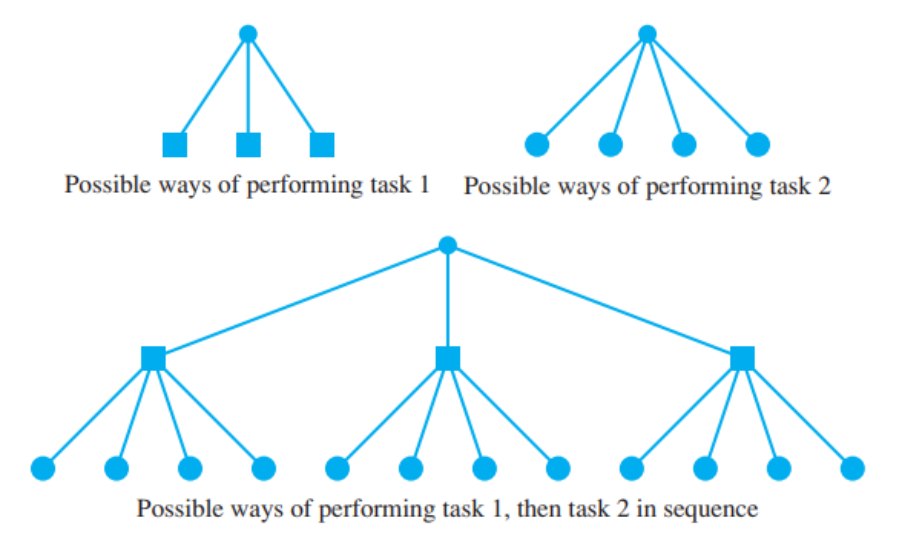
การเรียงสับเปลี่ยน

Created by : Rawesak Tanawongsuwan
rawesak.tan@mahidol.ac.th

Permutations

Theorem 1: Suppose that two tasks T_1 and T_2 are to be performed in sequence. If T_1 can be performed in n_1 ways, and for each of these ways T_2 can be performed in n_2 ways, then the sequence T_1T_2 can be performed in n_1n_2 ways.

(the multiplication principle of counting)



Permutations

Theorem 2: Suppose that tasks $T_1, T_2, \ldots, T_k$ are to be performed in sequence. If T_1 can be performed in n_1 ways, and for each of these ways T_2 can be performed in n_2 ways, and for each of these $n_1 n_2$ ways of performing $T_1 T_2$ in sequence, T_3 can be performed in n_3 ways, and so on, then the sequence $T_1 T_2 \ldots T_k$ can be performed in exactly $n_1 n_2 \ldots n_k$ ways.

Example 1

A label identifier, for a computer system, consists of one letter followed by three digits. If repetitions are allowed, how many distinct label identifiers are possible?

Solution

Letter	Digit	Digit	Digit

There are 26 possibilities for the beginning letter and there are 10 possibilities for each of the three digits. Thus, by the extended multiplication principle, there are 26 x 10 x 10 x 10 or 26,000 possible label identifiers.

Example 2

Let A be a set with n elements. How many subsets does A have?

Solution

We know from the set section that each subset of A is determined by its characteristic function, and if A has n elements, this function may be described as an array of 0's and 1's having length n. The first element of the array can be filled in two ways (with a 0 or a 1), and this is true for all succeeding elements as well.

0	1	1	0	0	…	1	1	0

Thus, by the extended multiplication principle, there are

$$\underbrace{2 * 2 * \cdots * 2}_{n} = 2^n$$

ways of filling the array, and therefore 2^n subsets of A.

Counting problem 1

How many different sequences, each of length r, can be formed using elements from A if

(a) elements in the sequence may be repeated?

(b) all elements in the sequence must be distinct?

Counting problem 1

How many different sequences, each of length r, can be formed using elements from A if

(a) elements in the sequence may be repeated?

(b) all elements in the sequence must be distinct?

			…		
Box 1	Box 2	Box 3		Box $r-1$	Box r

Let T_1 be the task "fill box 1", T_2 be the task "fill box 2", and so on.

The combined task $T_1 T_2 \ldots T_r$ represents the formation of the sequence.

Counting problem 1

- **Case (a) elements in the sequence may be repeated?**

			…		
Box 1	Box 2	Box 3		Box $r-1$	Box r

T_1 can be accomplished in n ways.

$T_2, T_3, \ldots, T_r$ can be accomplished in n ways too.

Then by the extended multiplication principle, the number of sequences

that can be formed is

$$\underbrace{\mathrm{n} * n * \cdots * n}_{r} = n^r$$

Counting problem 1

- **Case (b) all elements in the sequence must be distinct?**

			…		
Box 1	Box 2	Box 3		Box $r-1$	Box r

T_1 can be accomplished in n ways.

$T_2, T_3, \ldots, T_r$ can be accomplished in $n-1,\ n-2,\ \ldots,\ n-r+1$ ways.

Then by the extended multiplication principle, the number of sequences that can be formed is

$$n(n-1)(n-2)\ldots(n-r+1) = \frac{n!}{(n-r)!}$$

Theorem 3

- Let A be a set with n elements and $1 \leq r \leq n$. Then the number of sequences of length r that can be formed from elements of A, allowing repetitions, is n^r.

Example : How many three-letter words can be formed from letters in the set $\{a, b, y, z\}$ if repeated letters are allowed?

Solution : $n = 4, \quad \mathrm{r} = 3$, the number of words is $4^3 = 64$.

Theorem 4

- If $1 \leq r \leq n$, then the number of permutations of n objects taken r at a time $P(n,r) = n * (n-1) * (n-2) * \cdots * (n-r+1) = \frac{n!}{(n-r)!}$.

Example 1 : Let $A = \{1,2,3,4\}$.

The number of permutations of A taken 3 at a time $P(4,3) = 4 * 3 * 2 = 24$.

The number of permutations of A taken 2 at a time $P(4,2) = 4 * 3 = 12$.

The number of permutations of A $P(4,4) = 4! = 4 * 3 * 2 * 1 = 24$.

Example 2 : How many words of three distinct letters can be formed from the letters of he word **MUST** ?

The number if $P(4,3) = \frac{4!}{(4-3)!} = 24$

Dealing cards example

- Let A consist of all 52 cards in an ordinary deck of playing cards. Suppose that this deck is shuffled and a hand of five cards is dealt. A list of cards in this hand, in the order in which they were dealt, is a permutation of A taken five at a time.
- Examples would include AH, 3D, 5C, 2H, JS; 2H, 3H, 5H, QH, KD;

  JH, JD, JS, 4H, 4C; 3D, 2H, AH, JS, 5C.
- Note that the first and last hands are the same, but they represent different permutations since they were dealt in a different order.
- The number of permutations of A taken five at a time is

$$P(52,5) = \frac{52!}{47!} = 52 * 51 * 50 * 49 * 48 = 311{,}875{,}200.$$

- This is the number of five-card hands that can be dealt if we consider the order in which they were dealt.

Theorem 5

- The number of distinguishable permutations that can be formed from a collection of n objects where the first object appears k_1 times, the second object k_2 times, and so on, is

$$\frac{n!}{k_1!k_2!\cdots k_t!} \quad \text{where } k_1 + k_2 + \cdots + k_t = n$$

- Examples : The number of distinguishable words that can be formed from the letters of

BANANA = $\frac{6!}{3!1!2!} = 60.$

MISSISSIPPI = $\frac{11!}{1!4!4!2!} = 34{,}650$